

Housing Supply Elasticity and Transit Welfare: A Replication and Extension of Severen (2021)

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Abstract

Severen (2021) estimates the general-equilibrium welfare effects of the Los Angeles Metro Rail network. The paper finds commuting flows rose 10–20 % between station-adjacent census-tract pairs and derives a closed-city welfare gain of approximately \$93.55 per person per year, holding the city-wide housing supply elasticity ψ fixed at 1.602. We replicate the full Severen pipeline and extend the analysis in two directions. First, we attempt to estimate ψ separately for geographic sub-regions of the LA metro area (by coastal proximity, density quartile, and county group), motivated by the substantial spatial variation in local housing markets. The Bartik shift-share instrument proves too weak for sub-regional identification: first-stage F -statistics fall well below 5 in every partition tried, against a full-sample borderline of $F \approx 9$. Second, unable to obtain reliable sub-regional estimates, we vary ψ over a wide range in the calibrated GE model. We find the welfare estimate is essentially invariant to ψ : across a $5\times$ range (0.75–4.0), welfare moves by only \$0.33 per person (0.35%). This is expected for a small intervention in a car-dominated city, where the transit-induced demand shock near stations is too modest for housing supply conditions to affect aggregate welfare. "Faster-transit" counterfactuals confirm that reasonable improvements in train speed have limited impact: doubling the commuting disutility reduction associated with Metro connectivity adds only \$105/person/year on welfare.

Keywords: transit infrastructure, general equilibrium, housing supply elasticity, welfare analysis, Los Angeles

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1 Introduction

Urban transit investment is among the most capital-intensive interventions available to city governments. Evaluating whether such investments are welfare-improving requires a framework that accounts not only for the direct benefits to commuters but also for the general-equilibrium adjustments in labour and housing markets that follow. Severen (2021) provides one of the most complete such evaluations to date, applying a quantitative spatial model to the 1990–2002 build-out of the Los Angeles Metro Rail network.

The paper treats the housing supply elasticity ψ as a single city-wide constant, estimated at 1.602 using a Bartik shift-share instrument. This is a natural baseline, but it is a strong assumption: housing markets in the five-county LA metro area vary enormously across coastal and inland sub-markets, across density gradients from dense urban cores to low-density suburbs, and across counties with very different zoning regimes. If ψ varies meaningfully across the city, and if Metro Rail stations happen to be located in areas with systematically different supply conditions, the aggregate welfare estimate could mask important heterogeneity in who captures the gains from transit investment.

Our starting hypothesis was that housing supply conditions are a first-order determinant of the incidence of transit benefits. Near stations with inelastic supply, demand increases are capitalised into land prices and captured by existing landowners; near stations with elastic supply, housing quantities adjust and the benefits are spread more broadly. Understanding which regime prevails matters for both efficiency and distributional assessments of transit policy.

We extend the Severen (2021) replication in two directions. The first is an attempt to estimate ψ separately for geographic sub-regions of the LA metro area, splitting the sample by coastal proximity, 1990 population density quartile, and county group. The second, motivated by the failure of the first, is a systematic sensitivity analysis varying ψ across a wide range in the calibrated GE model.

Our key findings are as follows. The Bartik instrument does not have enough variation within sub-regions to separately identify ψ : first-stage F -statistics fall well below 5 in every geographic partition we try. This finding also casts some doubt on the precision of the original paper’s pooled estimate, for which $F \approx 9$ is already borderline. Turning to the sensitivity analysis, we find that the welfare gain from Metro Rail is almost completely insensitive to ψ . Across a $5\times$ range of the supply elasticity (0.75 to 4.0), the welfare gain moves by only \$0.33 per person — a 0.35 % variation around the \$93.55 baseline. We show that this insensitivity is a direct consequence of the scale of the intervention: Metro Rail is a small shock in a car-dominated city, and the transit-induced demand increase near stations is too modest for housing supply conditions to determine how much surplus commuters capture.

We also present "faster-transit" counterfactuals showing that amplifying the estimated commuting disutility benefit of Metro connectivity raises welfare roughly linearly in the multiplier. At $2\times$ the baseline benefit, welfare rises by \$105 per person, but the gain is almost entirely capitalised into land prices at the small fraction of tracts near Metro stations. The binding constraint on Metro Rail’s welfare impact seems to be the network coverage—only 4% of tracts lie near stations—not the magnitude of the commuting benefit per connected pair.

The remainder of this paper is organised as follows. Section 2 describes the Severen (2021) framework. Section 3 describes our modifications and methods. Section 4 presents results. Section 5 concludes.

2 The Severen (2021) Framework

2.1 Setting and data

Severen (2021) studies the commuting, labour-market, and housing-market effects of the Los Angeles Metro Rail network, which opened in three phases: the Blue Line (downtown LA to Long Beach) in 1990, and the Green and Red Lines in 1993–2000. The analysis covers 2,552 census tracts in the five-county LA metropolitan area (Los Angeles, Orange, Riverside, San Bernardino, and Ventura counties) over the period 1990–2002.

The main data sources are: (i) the 1990 and 2000 decennial censuses for commuting flows between home and work tracts; (ii) the Longitudinal Employer-Household Dynamics (LEHD) database extended through 2015 for robustness; (iii) the Neighbourhood Change Database (NCDB) for pre-trend tests; and (iv) GIS-computed travel times between all $2,552 \times 2,552$ tract pairs under observed and counterfactual transit networks.

2.2 Commuting flow estimates

The paper estimates a difference-in-differences specification for bilateral commuting flows. The treatment indicator is an indicator for whether both the home tract and the work tract contain (or lie within 250 m or 500 m of) a Metro station. Identification rests on the parallel pre-trends assumption, which is supported by the absence of significant pre-1990 trends in wages or housing values for station-adjacent tracts relative to control tracts.

The preferred estimates show that flows between tract pairs where both tracts contain a station rose by 10.0 % (SE 0.038), stable across eight progressively tighter geographic samples and confirmed by a Poisson pseudo-maximum-likelihood (PPML) specification. An additional proximity ring within 250 m raises the combined estimated effect to approximately 17.6 %.

2.3 Structural elasticities

To translate the reduced-form flow estimates into welfare-relevant parameters, the paper estimates four structural elasticities using a Bartik shift-share IV strategy. The housing supply elasticity ψ is the most consequential. It is estimated by instrumenting changes in housing density ($\Delta \ln N_{it}$) with a shift-share demand shock based on the 1990 industry mix interacted with national industry-level employment growth. The preferred estimate is $\psi = 1.602$, derived from the IV regression of $\Delta \ln P_{it}$ on $\Delta \ln N_{it}$ (the coefficient being $1/\psi = 0.624$).

The first stage for the housing supply estimation has a Cragg–Donald F -statistic of 363 and a Kleibergen–Paap F -statistic of 240. An internal consistency test — whether ψ estimated from housing consumption and from residential land use are consistent — returns a p -value of 0.616, supporting the structural interpretation.

2.4 General-equilibrium welfare model

The welfare calculation uses the hat-algebra approach of Dekle et al. (2008), extended to a multi-location setting with endogenous commuting. Workers choose a home and work location by drawing Fréchet-distributed idiosyncratic preference shocks, giving rise to a gravity equation

for commuting flows:

$$\pi_{ij} \propto \left(\frac{w_j}{\tau_{ij}} \right)^\theta \cdot B_j, \quad (1)$$

where w_j is the wage at destination tract j , τ_{ij} is the bilateral commuting cost, B_j captures workplace amenities, and θ is the commuting elasticity (calibrated from the literature). The housing market clears at each tract with supply elasticity ψ . The model is calibrated to the observed 2000 LA equilibrium and asked: by how much does welfare change when the observed transit travel-time improvements are removed? The answer — solved by fixed-point iteration converging in 6–11 iterations — is the welfare gain attributable to Metro Rail. The baseline result is **\$93.55 per resident per year** (closed-city), the figure we take as our benchmark throughout.

3 Modifications and Methods

3.1 Motivation: heterogeneous housing supply

Severen (2021) estimates a single pooled ψ for the five-county LA metro area. The implicit assumption is that housing supply elasticity does not vary systematically with station proximity or with the geographic characteristics of the sub-market in which a station is located. This assumption is strong: coastal tracts face stricter development constraints than inland tracts; already-dense central-city tracts have less room for upward densification than low-density suburban tracts; and LA County faces very different zoning conditions than Riverside or San Bernardino counties.

If ψ is systematically lower near stations (e.g., because stations were placed in already-dense, supply-constrained areas), the paper’s aggregate welfare estimate understates the extent to which transit benefits are capitalised into land values rather than shared broadly. Understanding the distribution of ψ across LA’s sub-markets therefore matters for both efficiency and distributional assessments of the Metro Rail programme.

3.2 Attempted sub-regional estimation

We attempt to estimate ψ separately for three types of geographic sub-regions, using the same Bartik IV specification as the original paper:

- **Coastal vs. inland.** Tracts within 5 km of the Pacific coastline versus all other tracts (296 coastal tracts, $\approx 2,257$ inland). Coastal distances are computed from a 1:50 000 Natural Earth coastline using California Albers projection.
- **Density quartiles.** Tracts are ranked by 1990 homeowners per km² and split at the median (binary) and into quartiles. The prior is that already-dense tracts face lower supply elasticity than low-density tracts.
- **County groups.** A *core* group (Los Angeles and Orange counties: dense, heavily zoned) versus a *fringe* group (Riverside, San Bernardino, and Ventura counties: lower density and more developable land). Both groups are large and industrially diverse, so the Bartik instrument was expected to retain adequate within-group variation.

Each specification uses the same IV as the original paper (the leave-one-out Bartik employment shock z_{it}), restricts to the estimation sample ($\tau_{c_hval} = 0$), and weights by 1990 homeowners.

3.3 Identification failure: instrument weakness

Every sub-regional split encounters the same problem: the Bartik instrument loses power when the sample is partitioned. Table 2 reports the Kleibergen–Paap Wald F -statistic for each split. The full-sample borderline of $F \approx 9$ (proxied by the inland specification, which contains $\approx 93\%$ of tracts) falls to $F < 5$ in every other partition and to $F < 1$ for coastal tracts, Density Q1, and the fringe counties individually.

The root cause is that the Bartik instrument derives its power from cross-industry variation in employment growth interacted with the pre-period industry mix. When the sample is restricted to a geographic sub-area, the effective number of tracts drops substantially, reducing the cross-sectional variation available to identify the supply curve. The coastal split is the extreme case: with only 188 tracts, the instrument is essentially uninformative (KP $F = 0.18$).

This finding also carries an implication for the original paper: a pooled KP $F \approx 9$ is below the conventional threshold of 10, suggesting the baseline $\psi = 1.602$ estimate rests on weaker identification than the headline numbers imply.

3.4 GE sensitivity analysis

Since sub-regional estimation is infeasible, we vary ψ across a grid spanning 0.5 to 4.0, covering a range from near-inelastic (San Francisco-type) to highly elastic (Sun Belt-type), and re-solve the full Severen GE model at each grid point. All other parameters are held at the paper’s baseline values. At each ψ we verify that:

- the stability condition holds (largest eigenvalue of the transition matrix < 1);
- closed-city and open-city equilibria converge (typically in 6–11 iterations to a tolerance of 10^{-5}).

The eigenvalue ranges from 0.778 ($\psi = 0.5$) to 0.951 ($\psi = 4.0$), confirming a unique equilibrium throughout.

3.5 "Faster-transit" counterfactuals

As a complementary exercise, we ask how welfare would change if Metro Rail generated a larger commuting benefit than actually observed (e.g. through faster train technology).

The commuting effect enters the model through the pair-specific bilateral utility shifter D_{ij} . In the hat-algebra counterfactual framework, the change in D_{ij} is governed by the estimated commuting parameters λ_{00}^D and λ_{02}^D , recovered from a difference-in-differences regression of log bilateral commuting flows on binary connectivity indicators T_{ij}^{00} and T_{ij}^{02} —equal to one when both home tract i and workplace tract j lie within 500 m of a station of the respective Metro line—controlling for OD-pair, residence-tract-by-year, and workplace-tract-by-year fixed effects.

For "speed multiplier" $s \geq 1$, the counterfactual adjusts D_{ij} as:

$$\hat{D}_{ij} = \exp\left(\lambda_{00}^D (s - 1) T_{ij}^{00} + \lambda_{02}^D (s - 1) T_{ij}^{02}\right), \quad (2)$$

where $\lambda_{00}^D = 0.149$ and $\lambda_{02}^D = 0.128$. Since both parameters are positive, $\hat{D}_{ij} > 1$ for Metro-connected pairs at $s > 1$, increasing their pair-specific commuting utility. At $s = 1$ the expression equals one (no change); at $s = 2$, each connected pair receives $(s - 1) = 1$ additional unit of the

estimated commuting utility gain—doubling the improvement attributed to Metro. For OD pairs where neither tract is near a station ($T_{ij}^{00} = T_{ij}^{02} = 0$, covering 96 % of pairs), $\hat{D}_{ij} = 1$ regardless of s .

It is worth noting that every Metro-connected pair receives the same proportional utility shift, independent of actual trip length or actual travel time, because the estimated λ^D parameters capture a flat average treatment effect. The GE model is re-solved in full at each s , so downstream price, wage, and population adjustments are endogenous. This counterfactual does not really test running trains run at a physically higher speed; it asks what equilibrium would result if Metro connectivity raised commuting utility by s times its estimated magnitude.

Starting from the 2000 calibrated baseline, we scale the structural commuting disutility parameters ($s - 1$) additional units of the estimated benefit to all Metro-connected origin-destination pairs. At $s = 2$ the commuting disutility reduction is twice its estimated magnitude. The GE model is re-solved at each multiplier and the incremental welfare gain on top of the \$93.55 baseline is recorded.

Re-solving the GE model across multipliers $s \in \{1.25, 1.5, 2, 3\}$ yields linearly increasing welfare gains of \$25–\$223 per person. The gain is almost entirely capitalised into land prices at the small number of station tracts. The exercise shows that the binding constraint on Metro Rail’s welfare impact is network coverage—only 94 of 2,552 tracts (4 %) lie near a station—rather than the magnitude of the commuting benefit per connected pair.

4 Results

4.1 Welfare is insensitive to ψ

Table 1 reports the closed-city welfare gain from Metro Rail across the ψ grid. The headline finding is that welfare is essentially invariant to ψ . Moving from $\psi = 0.75$ to $\psi = 4.0$ — a more than $5\times$ increase — reduces the welfare estimate by only \$0.33 per person, from \$93.71 to \$93.38. This represents a 0.35 % variation around the paper’s baseline of \$93.55. Over the full grid (0.5 to 4.0), the variation is \$0.41.

The relationship is monotone and negative: welfare is slightly *higher* under more inelastic supply ($\psi = 0.5$: \$93.79) than under more elastic supply ($\psi = 4.0$: \$93.38). The intuition is that with inelastic supply, housing prices rise more near stations, increasing the asset values captured by incumbent landowners. In the closed-city model this is a transfer within the city rather than a deadweight loss, so aggregate welfare is slightly higher. The open-city population gain varies more (0.081 % to 0.102 %), reflecting the composition effect: more elastic supply allows more in-migration, diluting per-capita welfare.

Table 1: Welfare sensitivity to housing supply elasticity (ψ)

ψ	Closed-city welfare (\$/person)	% of income	Open-city pop. gain
0.50	\$93.79	0.0442%	0.102%
0.75	\$93.71	0.0441%	0.097%
1.00	\$93.65	0.0441%	0.093%
1.25	\$93.60	0.0441%	0.091%
1.50	\$93.56	0.0440%	0.089%
1.602 (baseline)	\$93.55	0.0440%	0.088%
1.75	\$93.53	0.0440%	0.087%
2.00	\$93.50	0.0440%	0.086%
2.50	\$93.46	0.0440%	0.084%
3.00	\$93.43	0.0440%	0.083%
4.00	\$93.38	0.0440%	0.081%

GE model re-solved at each ψ value; all other parameters held at Severen (2021) baseline. Baseline row in bold. % of income computed relative to 2000 median individual earnings in LA metro area. Source: `run_welfare_psi_sensitivity.R`.

4.2 Sub-regional ψ estimates: instrument weakness

Table 2 reports the sub-regional ψ point estimates and first-stage F -statistics. The pattern is unambiguous: every geographic partition that reduces the sample substantially pushes the Kleibergen–Paap F -statistic below the conventional threshold of 10, and several produce $F < 1$. The coastal split, with only 188 tracts, is completely uninformative ($F = 0.18$); the ψ point estimate of 1.49 has a standard error of 5.72 and is statistically indistinguishable from any value. Even the county-group splits, designed to preserve within-group industry diversity, produce F -statistics of 3.24 (core) and 1.93 (fringe).

The most reliable sub-regional estimate is the inland specification ($\hat{\psi} = 0.49$, SE 0.17, KP $F = 9.89$), which proxies the full sample because 93 % of tracts are inland. Notably, this estimate is substantially *below* the paper’s pooled $\psi = 1.602$, and the density-quartile specifications where the instrument is marginally adequate (Q2, Q4) give estimates in the range 0.54–0.59, all below 1.602. If the reliable sub-regional estimates are taken at face value, the true ψ may be lower than the paper assumes, which would make the welfare estimate even less sensitive to the supply side.

Table 2: Sub-regional housing supply elasticity (ψ) estimates

Sub-region	<i>N</i> tracts	$\hat{\psi}$	SE	KP <i>F</i>
Full sample (pooled, Severen)	≈2,442	1.602	0.177	≈9
Core: LA + Orange Co.	≈1,850	0.170	0.097	3.24
Fringe: Riverside + SB + Ventura	≈590	−0.650	0.489	1.93
Coastal (<5 km Pacific)	188	1.488	5.716	0.18
Inland (≥5 km Pacific)	≈2,254	0.486	0.169	9.89
High density (≥ median, 1990)	≈1,200	0.435	0.174	1.83
Low density (< median, 1990)	≈1,200	0.493	0.400	7.46
Density Q1 (least dense)	≈600	0.088	1.080	0.007
Density Q2	≈600	0.590	0.311	6.48
Density Q3	≈600	0.367	0.240	2.58
Density Q4 (most dense)	≈600	0.540	0.255	6.47

Each row estimates ψ by IV (ivreg2, Bartik instrument, weighted by 1990 homeowners). KP *F* = Kleibergen–Paap Wald *F*-statistic. Conventional threshold for adequate identification: $F \geq 10$. Negative estimates for fringe counties and individual county splits reflect near-zero first stages and should not be given structural interpretation. Full-sample estimate reproduced from Severen (2021), Table 5a. Source: tracts_elasticities_coastal.do, tracts_elasticities_density.do, tracts_elasticities_countygroup.do.

4.3 "Faster-transit" counterfactuals

Table 3 reports the welfare gains from making Metro Rail trains "faster", expressed as the incremental gain on top of the \$93.55 baseline from current transit. The relationship is roughly linear for moderate speed improvements: doubling the reduction is disutility ($s = 2\times$) adds \$105/person/year, approximately four times the gain from a 25 % speed improvement (\$25/person).

Table 3: Welfare gains from faster Metro Rail trains

"Speed" multiplier	Incremental gain (\$/person)	Total welfare (\$/person)	Open-city pop. gain
1.0× (current)	\$0	\$93.55	0.088%
1.25×	\$25.17	\$118.72	0.024%
1.50×	\$51.09	\$144.64	0.048%
2.00×	\$105.26	\$198.81	0.099%
3.00×	\$223.47	\$317.02	0.211%
20.0×	\$5,921.70	\$6,015.25	—

Incremental gain = additional welfare on top of current transit (\$93.55/person). Multiplier s applies an additional disutility reduction of $(s - 1) \times$ the observed reduction on Metro corridors.

GE model converges at all levels of s .

Source: simulate_faster_transit.R; 20× from map_faster_transit_20x.R.

The 20× counterfactual is extreme but revealing. Welfare increase reaches \$5,922 per person per year, roughly 65 times the current transit benefit and approximately 20 % of median LA

household income. It would take imaginary transport technology (teleport?) to make a significant, non-linear impact on the GE outcomes.

4.4 The mechanism: why ψ does not matter

The insensitivity of welfare to ψ follows from the scale of the intervention. In the hat-algebra GE framework, the welfare gain from transit depends on the size of the commuting cost reduction, which in turn determines how much demand shifts toward station-adjacent tracts. LA Metro Rail induces roughly a 10–20 % increase in commuting flows between station pairs, but those station pairs represent a small fraction of all $2,552 \times 2,552$ possible tract-pair combinations. The aggregate demand shock to housing near stations is therefore small relative to the total LA housing stock.

Monocentric city simulations (using actual LA tract geography and an Alonso–Muth–Mills demand structure) confirm the mechanism. When the demand shock is small, the supply elasticity determines the price-vs-quantity split of the adjustment but leaves aggregate surplus nearly unchanged:

$$\Delta \ln P_n = \frac{D_n}{1 + \eta \psi}, \quad \Delta \ln N_n = \frac{\psi D_n}{1 + \eta \psi}, \quad (3)$$

where D_n is the demand shock at tract n and η is the demand elasticity. Total surplus equals $D_n \cdot (\Delta \ln P_n + \Delta \ln N_n)$ which is $D_n^2(1 + \psi)/(1 + \eta \psi)$ — a function of ψ , but one that varies by only 0.35 % across the relevant range when D_n is small. Only when the demand shock is large (as in the $20\times$ faster-transit case) does the price-vs-quantity split begin to affect aggregate welfare meaningfully.

5 Conclusion

This paper replicates the Severen (2021) analysis of LA Metro Rail welfare and extends it by asking whether the city-wide assumption on housing supply elasticity affects the results. Our answer is that it does not, and we show why.

The attempt to estimate ψ sub-regionally reveals a data limitation that the original paper does not fully acknowledge: the Bartik instrument, already borderline at the pooled level ($KP F \approx 9$), loses power entirely when the sample is partitioned by geography or density. This limits what can be learned about heterogeneity in housing supply conditions across LA’s sub-markets from this instrument and this dataset.

The GE sensitivity analysis, however, shows that this limitation is largely inconsequential for the paper’s headline welfare estimate. Across a $5\times$ range of ψ , welfare moves by \$0.33 — 0.35 % of the \$93.55 baseline. The mechanism is transparent: Metro Rail is a small intervention in a large, car-dominated city. The induced demand shock near stations is too modest for housing supply conditions to determine how much surplus commuters capture.

The "faster-transit" counterfactuals add a policy implication. Even doubling the reduction in commuting disutility would add only \$105/person/year showing linear increase. At an imaginary $20\times$ reduction the gain reaches \$5,922, but is almost entirely capitalised into land prices at the small fraction of station-adjacent tracts.

Taken together, these findings reinforce a central message of Severen (2021). The LA Metro Rail’s welfare impact is modest because the city’s commuting patterns are fundamentally shaped

by car infrastructure, and a limited rail network serving 4 % of census tracts cannot substantially reshape spatial equilibrium. Housing supply elasticity, however it varies across the metro area, is not the lever that would change this conclusion.

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